

- Q.25 Derive the formula of Sodium carbonate and Aluminium chloride.
- Q.26 Explain the formation of bond in O₂ molecule.
- Q.27 Write a note on Reverse osmosis.
- Q.28 Differentiate between Sludge and Scale.
- Q.29 State faraday first and second law of electrolysis.
- Q.30 What is Lead acid battery?
- Q.31 What are the qualities of a drinking water?
- Q.32 What is an ionic bond? Explain with an example.
- Q.33 What are the disadvantages of hard water in domestic use?
- Q.34 Explain Rutherford Model of atom.
- Q.35 State and explain Hund's Rule?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Write one example of alkane, alkene, alcohol, aldehyde, ketone with structure.
- Q.37 Explain dry cell with diagram.
- Q.38 a) Differentiate between sigma and pie bond.
b) Differentiate between compound and mixture

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Subject:- Applied Chemistry- I

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 In solids atoms have
a) Less space b) More space
c) no space d) none
- Q.2 pH of acid is _____ Than 7.
a) Less b) more
c) both d) none
- Q.3 Cause of hardness is
a) Bicarbonate ion b) chloride ion
c) both d) None
- Q.4 The valancy of Calcium is
a) 0 b) 1
c) 2 d) None

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- Q.5 Atomic mass is sum of _____ + _____
 a) neutron b) neutron + proton
 c) Proton d) None
- Q.6 Nitrogen molecule has
 a) triple bond b) single bond
 c) double bond d) None
- Q.7 Anions carry _____ charge
 a) Negative charge b) Both
 c) Positive Charge d) None
- Q.8 Temporary hardness of water is due to presence of Bicarbonate ion of
 a) Titanium
 b) calcium and magnesium
 c) both d) None
- Q.9 Strength is
 a) moles per unit volume
 b) Temp per unit volume
 c) Volume Independent
 d) Amount of solute per unit volume
- Q.10 Full form of TDS is
 a) Total Dissolved Solids
 b) Total Dissolved salts
 c) Both a & b d) None

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Examples of an electrolyte are _____ and _____
- Q.12 What is pollution?
- Q.13 Maximum number of electron in an atomic orbital is _____
- Q.14 Define the term solute.
- Q.15 Define Molarity of a solution.
- Q.16 1 Faraday = Coulomb.
- Q.17 Define boiler corrosion.
- Q.18 What is general formula of alkanes.
- Q.19 What is the formula of pH?
- Q.20 Define sigma bond.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define all units of hardness.
- Q.22 Write and balance any chemical equation.
- Q.23 Calculate the molarity of a solution prepared by dissolving 2g of NaOH in 500 ml of water. (Given Na=23, O=16, H=1)
- Q.24 What are the problems in boiler due to hard water?

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